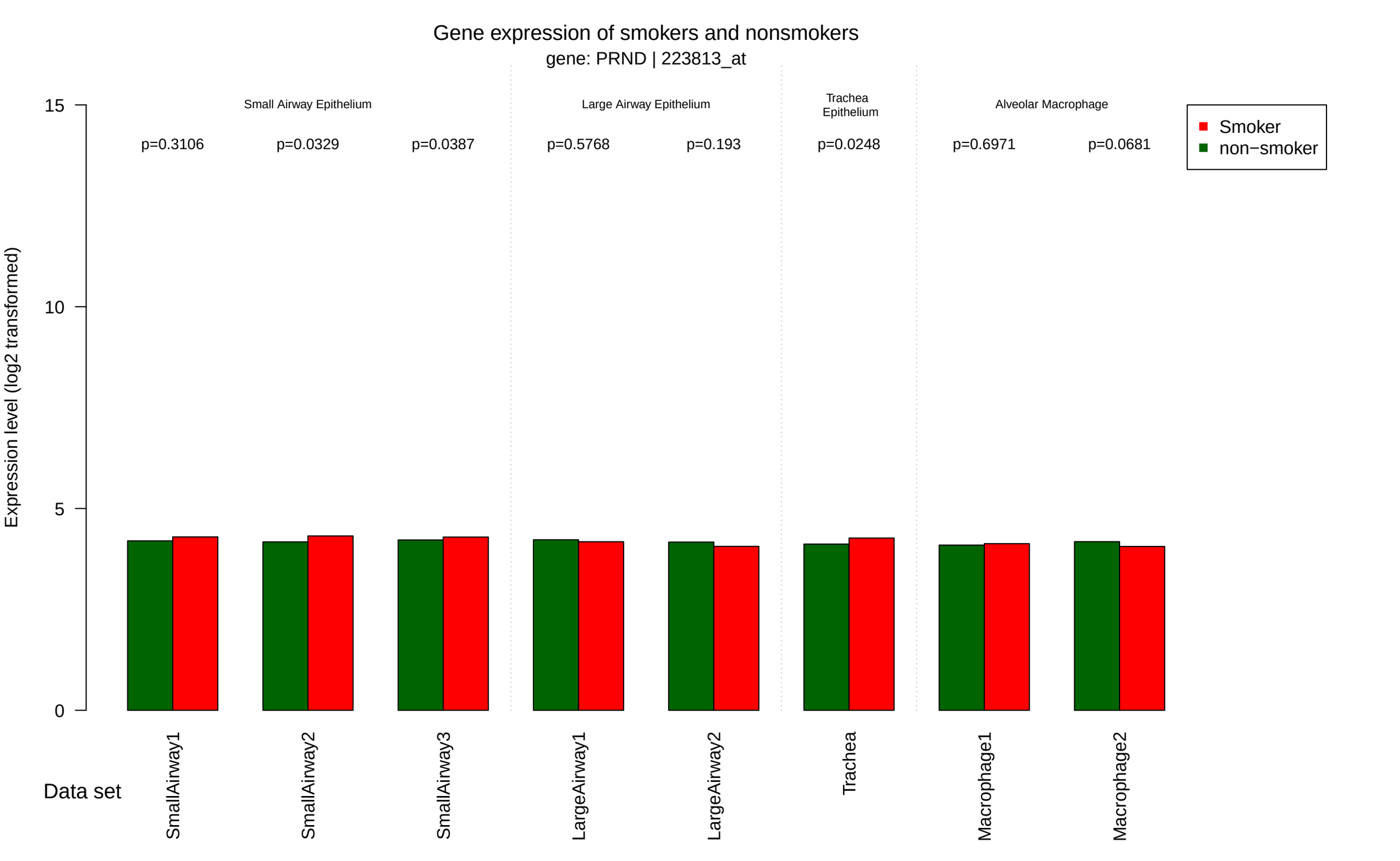
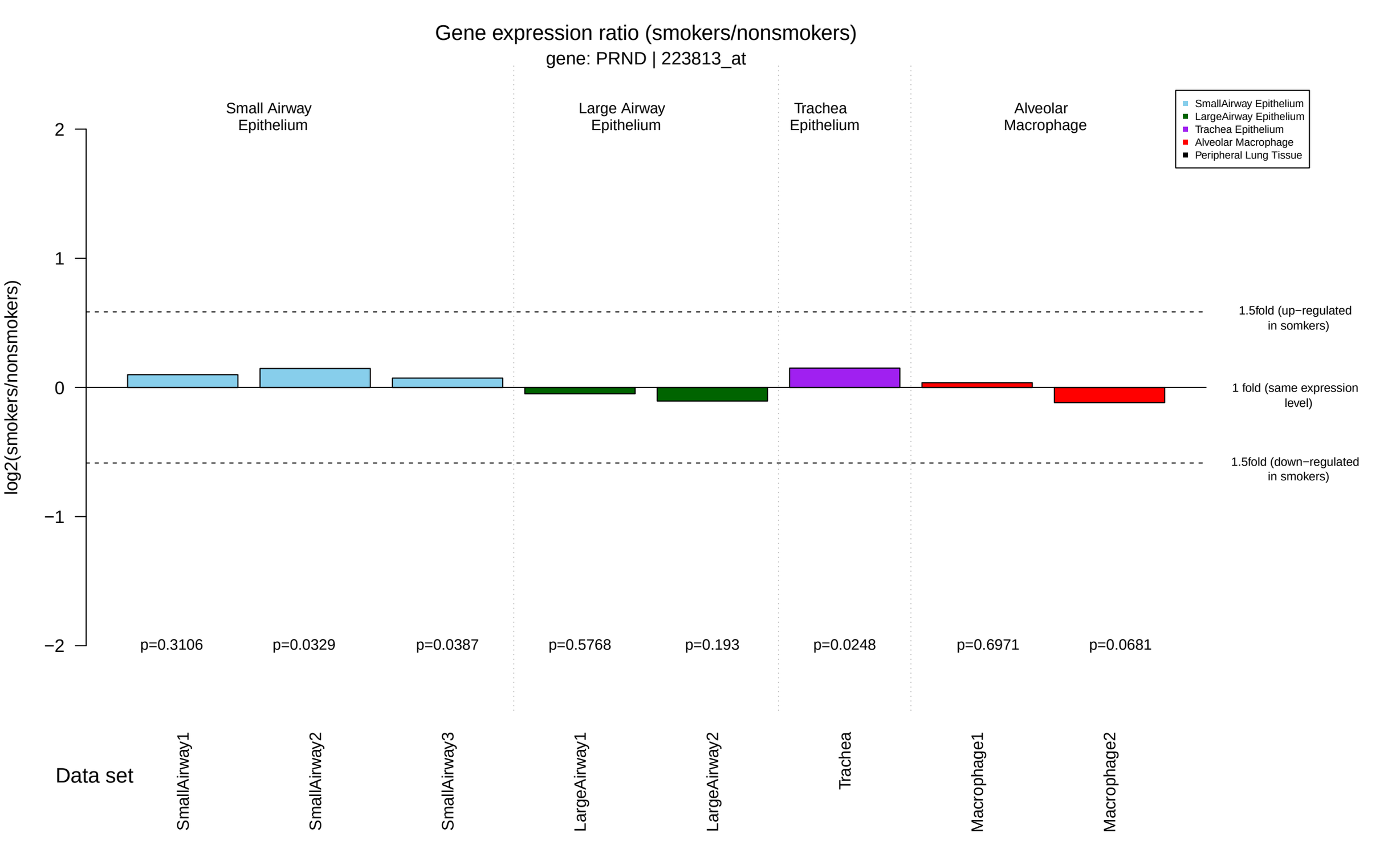
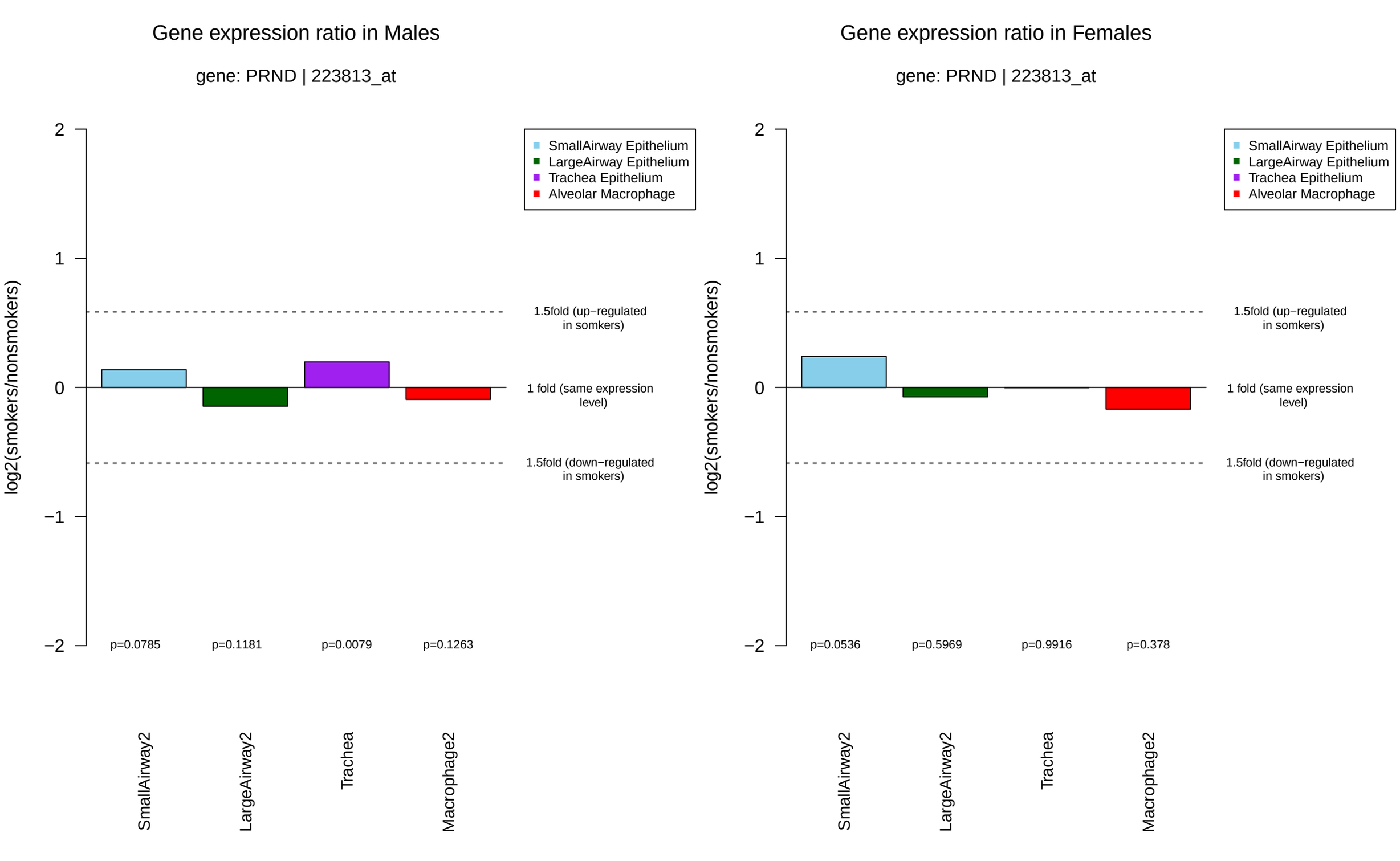
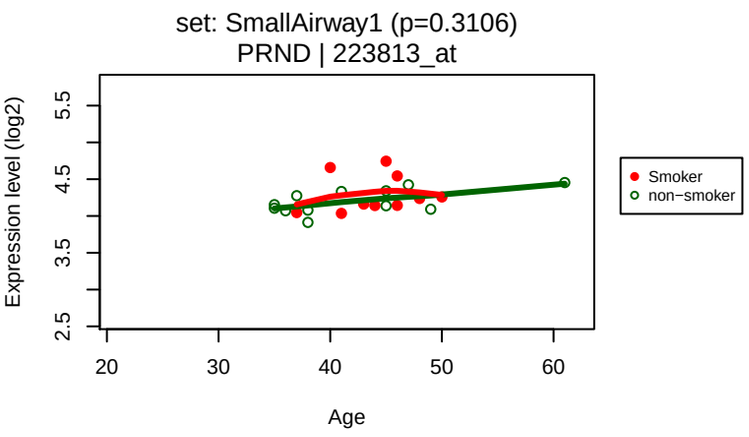
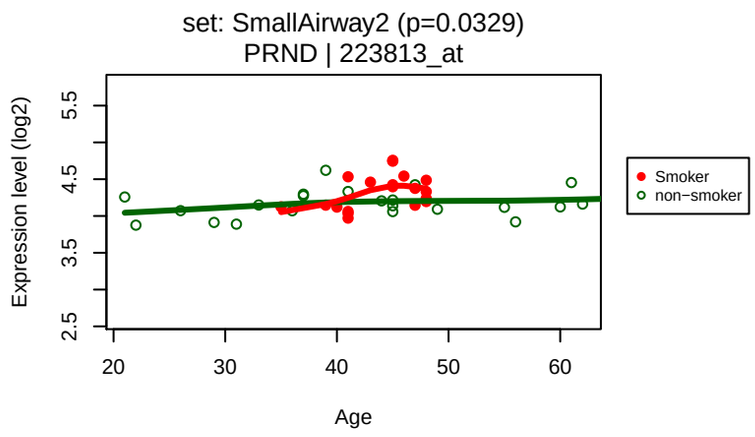
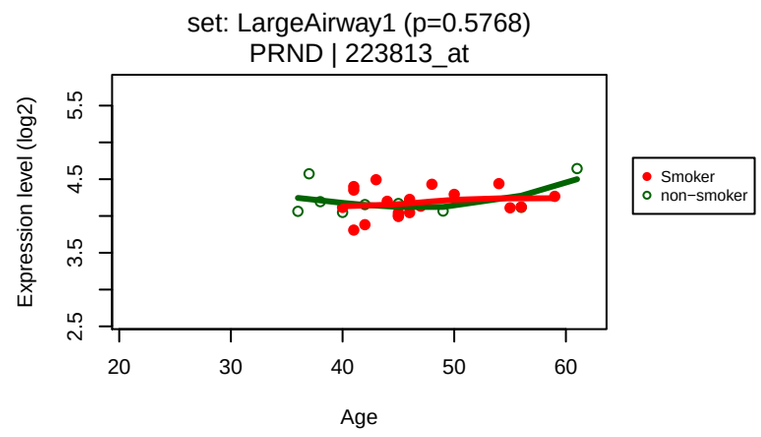

Gene expression of smokers and nonsmokers
gene: PRND | 223813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3106
p=0.0329
p=0.0387
p=0.5768
p=0.193
p=0.0248
p=0.6971
p=0.0681
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

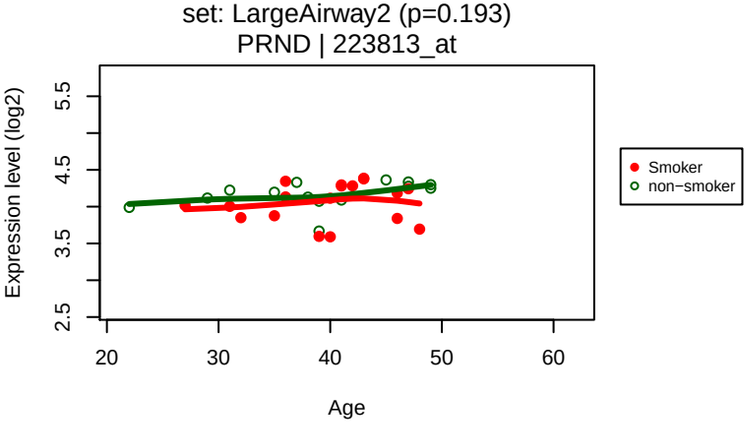

Gene expression ratio (smokers/nonsmokers)
gene: PRND | 223813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3106
p=0.0329
p=0.0387
p=0.5768
p=0.193
p=0.0248
p=0.6971
p=0.0681
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

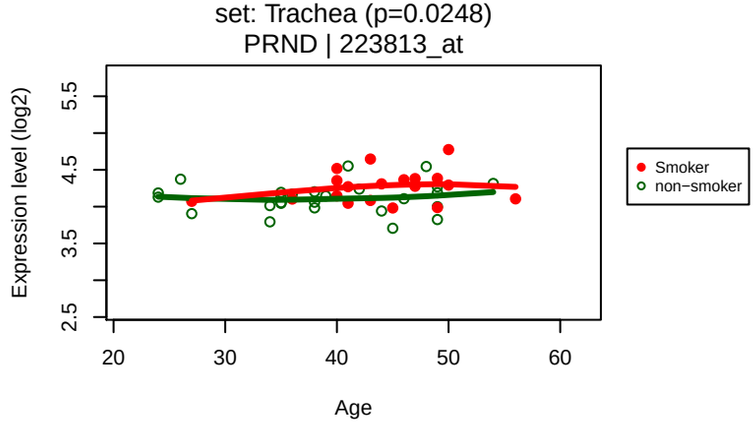

Gene expression ratio in Males
gene: PRND | 223813_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0785
p=0.1181
p=0.0079
p=0.1263
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRND | 223813_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0536
p=0.5969
p=0.9916
p=0.378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

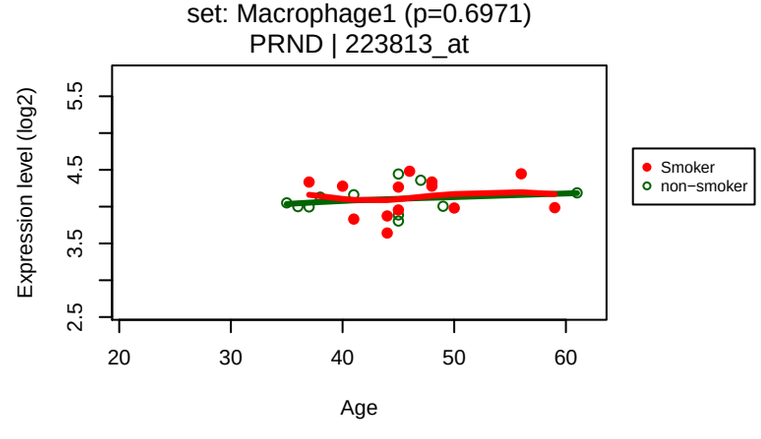

set: SmallAirway1 (p=0.3106)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker

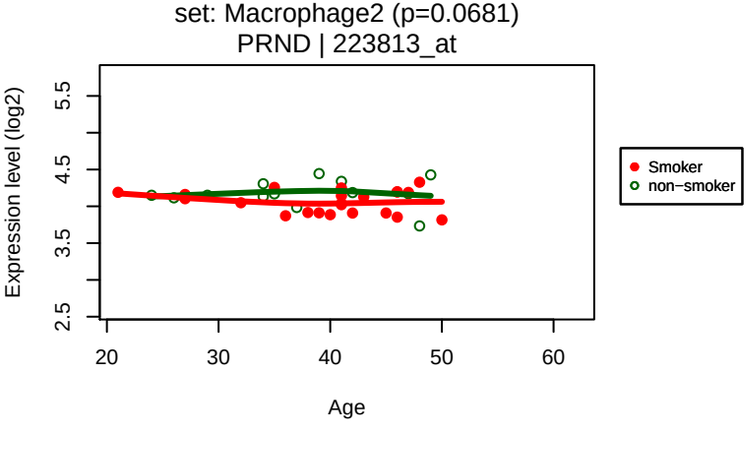

set: SmallAirway2 (p=0.0329)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker

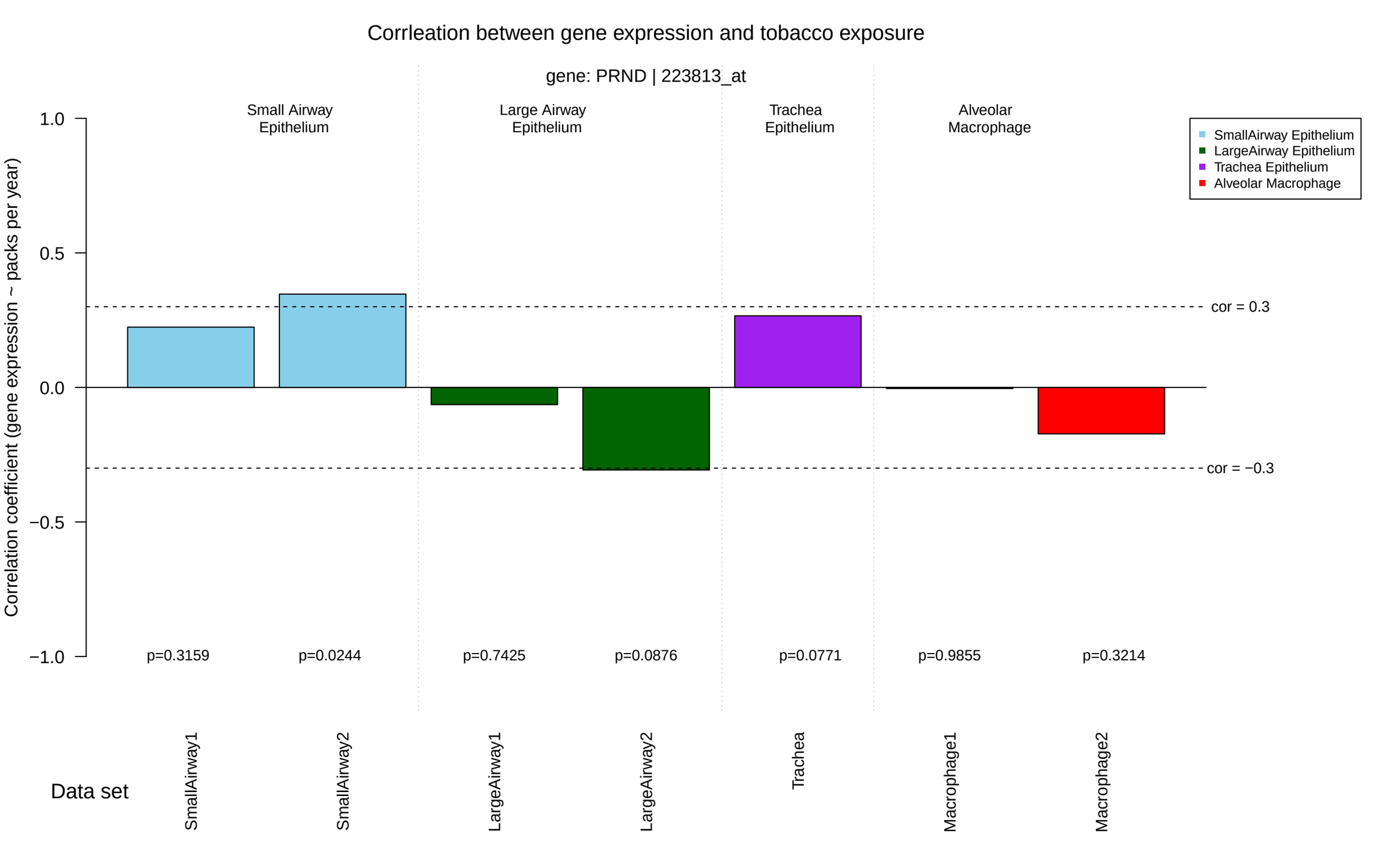

set: LargeAirway1 (p=0.5768)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.193)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0248)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6971)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0681)
PRND | 223813_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRND | 223813_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3159
p=0.0244
p=0.7425
p=0.0876
p=0.0771
p=0.9855
p=0.3214
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2